# Chapter 5

# VECTOR INTEGRATION

**ORDINARY INTEGRALS OF VECTORS.** Let $\mathbf{R}(u) = R_1(u)\mathbf{i} + R_2(u)\mathbf{j} + R_3(u)\mathbf{k}$ be a vector depending on a single scalar variable $u$, where $R_1(u)$, $R_2(u)$, $R_3(u)$ are supposed continuous in a specified interval. Then

$$\int \mathbf{R}(u)\,du = \mathbf{i}\int R_1(u)\,du + \mathbf{j}\int R_2(u)\,du + \mathbf{k}\int R_3(u)\,du$$

is called an *indefinite integral* of $\mathbf{R}(u)$. If there exists a vector $\mathbf{S}(u)$ such that $\mathbf{R}(u) = \frac{d}{du}(\mathbf{S}(u))$, then

$$\int \mathbf{R}(u)\,du = \int \frac{d}{du}(\mathbf{S}(u))\,du = \mathbf{S}(u) + \mathbf{c}$$

where $\mathbf{c}$ is an *arbitrary constant vector* independent of $u$. The *definite integral* between limits $u=a$ and $u=b$ can in such case be written

$$\int_a^b \mathbf{R}(u)\,du = \int_a^b \frac{d}{du}(\mathbf{S}(u))\,du = \mathbf{S}(u) + \mathbf{c}\,\Big|_a^b = \mathbf{S}(b) - \mathbf{S}(a)$$

This integral can also be defined as a limit of a sum in a manner analogous to that of elementary integral calculus.

**LINE INTEGRALS.** Let $\mathbf{r}(u) = x(u)\mathbf{i} + y(u)\mathbf{j} + z(u)\mathbf{k}$, where $\mathbf{r}(u)$ is the position vector of $(x,y,z)$, define a curve $C$ joining points $P_1$ and $P_2$, where $u=u_1$ and $u=u_2$ respectively.

We assume that $C$ is composed of a finite number of curves for each of which $\mathbf{r}(u)$ has a continuous derivative. Let $\mathbf{A}(x,y,z) = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ be a vector function of position defined and continuous along $C$. Then the integral of the tangential component of $\mathbf{A}$ along $C$ from $P_1$ to $P_2$, written as

$$\int_{P_1}^{P_2} \mathbf{A}\cdot d\mathbf{r} = \int_C \mathbf{A}\cdot d\mathbf{r} = \int_C A_1\,dx + A_2\,dy + A_3\,dz$$

is an example of a *line integral*. If $\mathbf{A}$ is the force $\mathbf{F}$ on a particle moving along $C$, this line integral represents the work done by the force. If $C$ is a closed curve (which we shall suppose is a *simple closed curve*, i.e. a curve which does not intersect itself anywhere) the integral around $C$ is often denoted by

$$\oint \mathbf{A}\cdot d\mathbf{r} = \oint A_1\,dx + A_2\,dy + A_3\,dz$$

In aerodynamics and fluid mechanics this integral is called the *circulation* of $\mathbf{A}$ about $C$, where $\mathbf{A}$ represents the velocity of a fluid.

In general, any integral which is to be evaluated along a curve is called a line integral. Such integrals can be defined in terms of limits of sums as are the integrals of elementary calculus.

For methods of evaluation of line integrals, see the Solved Problems.

The following theorem is important.

**THEOREM.** If $\mathbf{A} = \nabla\phi$ everywhere in a region $R$ of space, defined by $a_1 \leqq x \leqq a_2$, $b_1 \leqq y \leqq b_2$, $c_1 \leqq z \leqq c_2$, where $\phi(x,y,z)$ is single-valued and has continuous derivatives in $R$, then

1. $\displaystyle\int_{P_1}^{P_2} \mathbf{A}\cdot d\mathbf{r}$ is independent of the path $C$ in $R$ joining $P_1$ and $P_2$.

2. $\displaystyle\oint_C \mathbf{A}\cdot d\mathbf{r} = 0$ around any closed curve $C$ in $R$.

In such case $\mathbf{A}$ is called a *conservative vector field* and $\phi$ is its *scalar potential.*

A vector field $\mathbf{A}$ is conservative if and only if $\nabla\times\mathbf{A} = \mathbf{0}$, or equivalently $\mathbf{A} = \nabla\phi$. In such case $\mathbf{A}\cdot d\mathbf{r} = A_1\,dx + A_2\,dy + A_3\,dz = d\phi$, an exact differential. See Problems 10-14.

**SURFACE INTEGRALS.** Let $S$ be a two-sided surface, such as shown in the figure below. Let one side of $S$ be considered arbitrarily as the positive side (if $S$ is a closed surface this is taken as the outer side). A unit normal $\mathbf{n}$ to any point of the positive side of $S$ is called a *positive* or *outward drawn* unit normal.

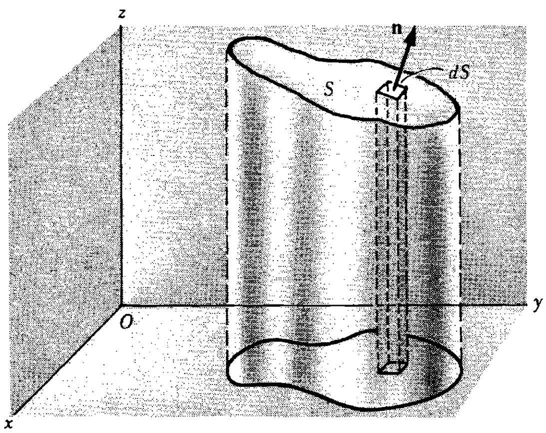
Associate with the differential of surface area $dS$ a vector $d\mathbf{S}$ whose magnitude is $dS$ and whose direction is that of $\mathbf{n}$. Then $d\mathbf{S} = \mathbf{n}\,dS$. The integral

$$\iint_S \mathbf{A}\cdot d\mathbf{S} = \iint_S \mathbf{A}\cdot\mathbf{n}\,dS$$

is an example of a surface integral called the *flux* of $\mathbf{A}$ over $S$. Other surface integrals are

$$\iint_S \phi\,dS, \quad \iint_S \phi\,\mathbf{n}\,dS, \quad \iint_S \mathbf{A}\times d\mathbf{S}$$

where $\phi$ is a scalar function. Such integrals can be defined in terms of limits of sums as in elementary calculus (see Problem 17).




The notation $\displaystyle\oiint_S$ is sometimes used to indicate integration over the closed surface $S$. Where no confusion can arise the notation $\displaystyle\oint_S$ may also be used.

To evaluate surface integrals, it is convenient to express them as double integrals taken over the projected area of the surface $S$ on one of the coordinate planes. This is possible if any line perpendicular to the coordinate plane chosen meets the surface in no more than one point. However, this does not pose any real problem since we can generally subdivide $S$ into surfaces which do satisfy this restriction.

**VOLUME INTEGRALS.** Consider a closed surface in space enclosing a volume $V$. Then

$$\iiint_V \mathbf{A}\,dV \quad \text{and} \quad \iiint_V \phi\,dV$$

are examples of *volume integrals* or *space integrals* as they are sometimes called. For evaluation of such integrals, see the Solved Problems.

## SOLVED PROBLEMS

1. If $\mathbf{R}(u) = (u-u^2)\mathbf{i} + 2u^3\,\mathbf{j} - 3\mathbf{k}$, find (*a*) $\int \mathbf{R}(u)\,du$ and (*b*) $\int_1^2 \mathbf{R}(u)\,du$.

(*a*)
$$\begin{aligned}\int \mathbf{R}(u)\,du &= \int [(u-u^2)\mathbf{i} + 2u^3\,\mathbf{j} - 3\mathbf{k}]\,du \\ &= \mathbf{i}\int (u-u^2)\,du + \mathbf{j}\int 2u^3\,du + \mathbf{k}\int -3\,du \\ &= \mathbf{i}\left(\frac{u^2}{2} - \frac{u^3}{3} + c_1\right) + \mathbf{j}\left(\frac{u^4}{2} + c_2\right) + \mathbf{k}(-3u + c_3) \\ &= \left(\frac{u^2}{2} - \frac{u^3}{3}\right)\mathbf{i} + \frac{u^4}{2}\mathbf{j} - 3u\,\mathbf{k} + c_1\,\mathbf{i} + c_2\,\mathbf{j} + c_3\,\mathbf{k} \\ &= \left(\frac{u^2}{2} - \frac{u^3}{3}\right)\mathbf{i} + \frac{u^4}{2}\mathbf{j} - 3u\,\mathbf{k} + \mathbf{c}\end{aligned}$$

where $\mathbf{c}$ is the constant vector $c_1\,\mathbf{i} + c_2\,\mathbf{j} + c_3\,\mathbf{k}$.

(*b*) From (*a*),
$$\begin{aligned}\int_1^2 \mathbf{R}(u)\,du &= \left(\frac{u^2}{2} - \frac{u^3}{3}\right)\mathbf{i} + \frac{u^4}{2}\mathbf{j} - 3u\,\mathbf{k} + \mathbf{c}\,\Big|_1^2 \\ &= \left[\left(\frac{2^2}{2} - \frac{2^3}{3}\right)\mathbf{i} + \frac{2^4}{2}\mathbf{j} - 3(2)\mathbf{k} + \mathbf{c}\right] - \left[\left(\frac{1^2}{2} - \frac{1^3}{3}\right)\mathbf{i} + \frac{1^4}{2}\mathbf{j} - 3(1)\mathbf{k} + \mathbf{c}\right] \\ &= -\frac{5}{6}\mathbf{i} + \frac{15}{2}\mathbf{j} - 3\mathbf{k}\end{aligned}$$

*Another Method.*

$$\begin{aligned}\int_1^2 \mathbf{R}(u)\,du &= \mathbf{i}\int_1^2 (u-u^2)du + \mathbf{j}\int_1^2 2u^3\,du + \mathbf{k}\int_1^2 -3\,du \\ &= \mathbf{i}\left(\frac{u^2}{2} - \frac{u^3}{3}\right)\Big|_1^2 + \mathbf{j}\left(\frac{u^4}{2}\right)\Big|_1^2 + \mathbf{k}(-3u)\Big|_1^2 = -\frac{5}{6}\mathbf{i} + \frac{15}{2}\mathbf{j} - 3\mathbf{k}\end{aligned}$$

2. The acceleration of a particle at any time $t \geq 0$ is given by
$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 12\cos 2t\,\mathbf{i} - 8\sin 2t\,\mathbf{j} + 16t\,\mathbf{k}$$
If the velocity $\mathbf{v}$ and displacement $\mathbf{r}$ are zero at $t=0$, find $\mathbf{v}$ and $\mathbf{r}$ at any time.

Integrating,
$$\begin{aligned}\mathbf{v} &= \mathbf{i}\int 12\cos 2t\,dt + \mathbf{j}\int -8\sin 2t\,dt + \mathbf{k}\int 16t\,dt \\ &= 6\sin 2t\,\mathbf{i} + 4\cos 2t\,\mathbf{j} + 8t^2\,\mathbf{k} + \mathbf{c}_1\end{aligned}$$

Putting $\mathbf{v}=\mathbf{0}$ when $t=0$, we find $\mathbf{0} = 0\mathbf{i} + 4\mathbf{j} + 0\mathbf{k} + \mathbf{c}_1$ and $\mathbf{c}_1 = -4\mathbf{j}$.

Then $\mathbf{v} = 6\sin 2t\,\mathbf{i} + (4\cos 2t - 4)\,\mathbf{j} + 8t^2\,\mathbf{k}$

so that $\dfrac{d\mathbf{r}}{dt} = 6\sin 2t\,\mathbf{i} + (4\cos 2t - 4)\,\mathbf{j} + 8t^2\,\mathbf{k}$.

Integrating,
$$\begin{aligned}\mathbf{r} &= \mathbf{i}\int 6\sin 2t\,dt + \mathbf{j}\int (4\cos 2t - 4)\,dt + \mathbf{k}\int 8t^2\,dt \\ &= -3\cos 2t\,\mathbf{i} + (2\sin 2t - 4t)\,\mathbf{j} + \frac{8}{3}t^3\,\mathbf{k} + \mathbf{c}_2\end{aligned}$$

Putting $\mathbf{r}=\mathbf{0}$ when $t=0$, $\mathbf{0} = -3\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} + \mathbf{c}_2$ and $\mathbf{c}_2 = 3\mathbf{i}$.

Then $\mathbf{r} = (3 - 3\cos 2t)\mathbf{i} + (2\sin 2t - 4t)\mathbf{j} + \frac{8}{3}t^3\mathbf{k}$.

3. Evaluate $\int \mathbf{A} \times \frac{d^2\mathbf{A}}{dt^2}\,dt$.

$$\frac{d}{dt}(\mathbf{A} \times \frac{d\mathbf{A}}{dt}) = \mathbf{A} \times \frac{d^2\mathbf{A}}{dt^2} + \frac{d\mathbf{A}}{dt} \times \frac{d\mathbf{A}}{dt} = \mathbf{A} \times \frac{d^2\mathbf{A}}{dt^2}$$

Integrating, $\int \mathbf{A} \times \frac{d^2\mathbf{A}}{dt^2}\,dt = \int \frac{d}{dt}(\mathbf{A} \times \frac{d\mathbf{A}}{dt})\,dt = \mathbf{A} \times \frac{d\mathbf{A}}{dt} + \mathbf{c}$.

4. The equation of motion of a particle $P$ of mass $m$ is given by

$$m\frac{d^2\mathbf{r}}{dt^2} = f(r)\,\mathbf{r}_1$$

where $\mathbf{r}$ is the position vector of $P$ measured from an origin $O$, $\mathbf{r}_1$ is a unit vector in the direction $\mathbf{r}$, and $f(r)$ is a function of the distance of $P$ from $O$.

(a) Show that $\mathbf{r} \times \frac{d\mathbf{r}}{dt} = \mathbf{c}$ where $\mathbf{c}$ is a constant vector.

(b) Interpret physically the cases $f(r) < 0$ and $f(r) > 0$.

(c) Interpret the result in (a) geometrically.

(d) Describe how the results obtained relate to the motion of the planets in our solar system.

(a) Multiply both sides of $m\frac{d^2\mathbf{r}}{dt^2} = f(r)\,\mathbf{r}_1$ by $\mathbf{r}\times$. Then

$$m\mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2} = f(r)\,\mathbf{r} \times \mathbf{r}_1 = \mathbf{0}$$

since $\mathbf{r}$ and $\mathbf{r}_1$ are collinear and so $\mathbf{r} \times \mathbf{r}_1 = \mathbf{0}$. Thus

$$\mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2} = \mathbf{0} \quad \text{and} \quad \frac{d}{dt}(\mathbf{r} \times \frac{d\mathbf{r}}{dt}) = \mathbf{0}$$

Integrating, $\mathbf{r} \times \frac{d\mathbf{r}}{dt} = \mathbf{c}$, where $\mathbf{c}$ is a constant vector. (Compare with Problem 3).

(b) If $f(r) < 0$ the acceleration $\frac{d^2\mathbf{r}}{dt^2}$ has direction opposite to $\mathbf{r}_1$; hence the force is directed toward $O$ and the particle is always *attracted* toward $O$.

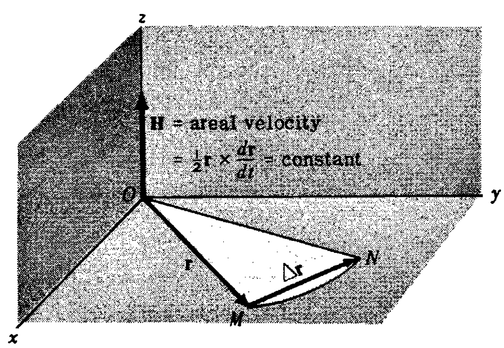
If $f(r) > 0$ the force is directed away from $O$ and the particle is under the influence of a *repulsive* force at $O$.

A force directed toward or away from a fixed point $O$ and having magnitude depending only on the distance $r$ from $O$ is called a *central force*.

(c) In time $\Delta t$ the particle moves from $M$ to $N$ (see adjoining figure). The area swept out by the position vector in this time is approximately half the area of a parallelogram with sides $\mathbf{r}$ and $\Delta\mathbf{r}$, or $\frac{1}{2}\mathbf{r} \times \Delta\mathbf{r}$. Then the approximate area swept out by the radius vector per unit time is $\frac{1}{2}\mathbf{r} \times \frac{\Delta\mathbf{r}}{\Delta t}$; hence the instantaneous time rate of change in area is

$$\lim_{\Delta t \to 0} \frac{1}{2}\mathbf{r} \times \frac{\Delta\mathbf{r}}{\Delta t} = \frac{1}{2}\mathbf{r} \times \frac{d\mathbf{r}}{dt} = \frac{1}{2}\mathbf{r} \times \mathbf{v}$$

where $\mathbf{v}$ is the instantaneous velocity of the parti-



cle. The quantity $\mathbf{H} = \frac{1}{2}\mathbf{r} \times \frac{d\mathbf{r}}{dt} = \frac{1}{2}\mathbf{r} \times \mathbf{v}$ is called the *areal velocity*. From part (*a*),

$$\text{Areal Velocity} = \mathbf{H} = \tfrac{1}{2}\mathbf{r} \times \frac{d\mathbf{r}}{dt} = \text{constant}$$

Since $\mathbf{r}\cdot\mathbf{H} = 0$, the motion takes place in a plane, which we take as the $xy$ plane in the figure above.

(*d*) A planet (such as the earth) is attracted toward the sun according to Newton's universal law of gravitation, which states that any two objects of mass $m$ and $M$ respectively are attracted toward each other with a force of magnitude $F = \frac{GMm}{r^2}$, where $r$ is the distance between objects and $G$ is a universal constant. Let $m$ and $M$ be the masses of the planet and sun respectively and choose a set of coordinate axes with the origin $O$ at the sun. Then the equation of motion of the planet is

$$m\frac{d^2\mathbf{r}}{dt^2} = -\frac{GMm}{r^2}\mathbf{r}_1 \qquad \text{or} \qquad \frac{d^2\mathbf{r}}{dt^2} = -\frac{GM}{r^2}\mathbf{r}_1$$

assuming the influence of the other planets to be negligible.

According to part (*c*), a planet moves around the sun so that its position vector sweeps out equal areas in equal times. This result and that of Problem 5 are two of Kepler's famous three laws which he deduced empirically from volumes of data compiled by the astronomer Tycho Brahe. These laws enabled Newton to formulate his universal law of gravitation. For Kepler's third law see Problem 36.

5. Show that the path of a planet around the sun is an ellipse with the sun at one focus.

From Problems 4(*c*) and 4(*d*),

$$(1) \qquad \frac{d\mathbf{v}}{dt} = -\frac{GM}{r^2}\mathbf{r}_1$$

$$(2) \qquad \mathbf{r}\times\mathbf{v} = 2\mathbf{H} = \mathbf{h}$$

Now $\mathbf{r} = r\,\mathbf{r}_1$, $\frac{d\mathbf{r}}{dt} = r\frac{d\mathbf{r}_1}{dt} + \frac{dr}{dt}\mathbf{r}_1$ so that

$$(3) \qquad \mathbf{h} = \mathbf{r}\times\mathbf{v} = r\,\mathbf{r}_1 \times \left(r\frac{d\mathbf{r}_1}{dt} + \frac{dr}{dt}\mathbf{r}_1\right) = r^2\mathbf{r}_1 \times \frac{d\mathbf{r}_1}{dt}$$

From (*1*), $\frac{d\mathbf{v}}{dt}\times\mathbf{h} = -\frac{GM}{r^2}\mathbf{r}_1\times\mathbf{h} = -GM\,\mathbf{r}_1\times\left(\mathbf{r}_1\times\frac{d\mathbf{r}_1}{dt}\right)$

$$= -GM\left[\left(\mathbf{r}_1\cdot\frac{d\mathbf{r}_1}{dt}\right)\mathbf{r}_1 - (\mathbf{r}_1\cdot\mathbf{r}_1)\frac{d\mathbf{r}_1}{dt}\right] = GM\frac{d\mathbf{r}_1}{dt}$$

using equation (*3*) and the fact that $\mathbf{r}_1\cdot\frac{d\mathbf{r}_1}{dt} = 0$ (Problem 9, Chapter 3).

But since $\mathbf{h}$ is a constant vector, $\frac{d\mathbf{v}}{dt}\times\mathbf{h} = \frac{d}{dt}(\mathbf{v}\times\mathbf{h})$ so that

$$\frac{d}{dt}(\mathbf{v}\times\mathbf{h}) = GM\frac{d\mathbf{r}_1}{dt}$$

Integrating, $$\mathbf{v}\times\mathbf{h} = GM\,\mathbf{r}_1 + \mathbf{p}$$

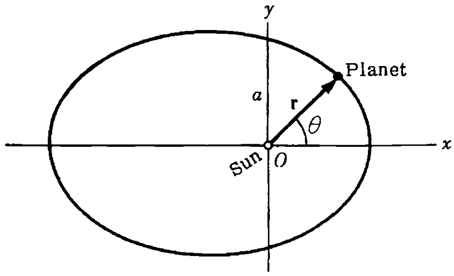
from which
$$\mathbf{r}\cdot(\mathbf{v}\times\mathbf{h}) = GM\,\mathbf{r}\cdot\mathbf{r}_1 + \mathbf{r}\cdot\mathbf{p}$$
$$= GM\,r + r\,\mathbf{r}_1\cdot\mathbf{p} = GM\,r + r\,p\cos\theta$$

where $\mathbf{p}$ is an arbitrary constant vector with magnitude $p$, and $\theta$ is the angle between $\mathbf{p}$ and $\mathbf{r}_1$.

Since $\mathbf{r}\cdot(\mathbf{v}\times\mathbf{h}) = (\mathbf{r}\times\mathbf{v})\cdot\mathbf{h} = \mathbf{h}\cdot\mathbf{h} = h^2$, we have $h^2 = GMr + r\,p\cos\theta$ and

$$r = \frac{h^2}{GM + p\cos\theta} = \frac{h^2/GM}{1 + (p/GM)\cos\theta}$$

From analytic geometry, the polar equation of a conic section with focus at the origin and eccentricity $\epsilon$ is $r = \dfrac{a}{1+\epsilon\cos\theta}$ where $a$ is a constant. Comparing this with the equation derived, it is seen that the required orbit is a conic section with eccentricity $\epsilon = p/GM$. The orbit is an ellipse, parabola or hyperbola according as $\epsilon$ is less than, equal to or greater than one. Since orbits of planets are closed curves it follows that they must be ellipses.




Ellipse $r = \dfrac{a}{1+\epsilon\cos\theta}$

## LINE INTEGRALS

6. If $\mathbf{A} = (3x^2+6y)\mathbf{i} - 14yz\,\mathbf{j} + 20xz^2\,\mathbf{k}$, evaluate $\displaystyle\int_C \mathbf{A}\cdot d\mathbf{r}$ from (0,0,0) to (1,1,1) along the following paths $C$:
   (a) $x=t,\ y=t^2,\ z=t^3$.
   (b) the straight lines from (0,0,0) to (1,0,0), then to (1,1,0), and then to (1,1,1).
   (c) the straight line joining (0,0,0) and (1,1,1).

$$\int_C \mathbf{A}\cdot d\mathbf{r} = \int_C [(3x^2+6y)\mathbf{i} - 14yz\,\mathbf{j} + 20xz^2\mathbf{k}]\cdot(dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k})$$
$$= \int_C (3x^2+6y)\,dx - 14yz\,dy + 20xz^2\,dz$$

(a) If $x=t,\ y=t^2,\ z=t^3$, points (0,0,0) and (1,1,1) correspond to $t=0$ and $t=1$ respectively. Then

$$\int_C \mathbf{A}\cdot d\mathbf{r} = \int_{t=0}^{1} (3t^2+6t^2)\,dt - 14(t^2)(t^3)\,d(t^2) + 20(t)(t^3)^2\,d(t^3)$$
$$= \int_{t=0}^{1} 9t^2\,dt - 28t^6\,dt + 60t^9\,dt$$
$$= \int_{t=0}^{1} (9t^2-28t^6+60t^9)\,dt = 3t^3 - 4t^7 + 6t^{10}\Big|_0^1 = 5$$

*Another Method.*

Along $C$, $\mathbf{A} = 9t^2\mathbf{i} - 14t^5\mathbf{j} + 20t^7\mathbf{k}$ and $\mathbf{r} = x\mathbf{i}+y\mathbf{j}+z\mathbf{k} = t\mathbf{i}+t^2\mathbf{j}+t^3\mathbf{k}$ and $d\mathbf{r} = (\mathbf{i}+2t\mathbf{j}+3t^2\mathbf{k})\,dt$.

$$\text{Then}\quad \int_C \mathbf{A}\cdot d\mathbf{r} = \int_{t=0}^{1} (9t^2\,\mathbf{i} - 14t^5\,\mathbf{j} + 20t^7\,\mathbf{k})\cdot(\mathbf{i} + 2t\,\mathbf{j} + 3t^2\,\mathbf{k})\,dt$$
$$= \int_0^1 (9t^2 - 28t^6 + 60t^9)\,dt = 5$$

(b) Along the straight line from (0,0,0) to (1,0,0) $y=0,\ z=0,\ dy=0,\ dz=0$ while $x$ varies from 0 to 1. Then the integral over this part of the path is

$$\int_{x=0}^{1} \big(3x^2+6(0)\big)\,dx - 14(0)(0)(0) + 20x(0)^2(0) = \int_{x=0}^{1} 3x^2\,dx = x^3\Big|_0^1 = 1$$

Along the straight line from (1,0,0) to (1,1,0) $x=1,\ z=0,\ dx=0,\ dz=0$ while $y$ varies from 0 to 1. Then the integral over this part of the path is

$$\int_{y=0}^{1} (3(1)^2+6y)0 - 14y(0)\,dy + 20(1)(0)^2\,0 = 0$$

Along the straight line from (1,1,0) to (1,1,1) $x=1$, $y=1$, $dx=0$, $dy=0$ while $z$ varies from 0 to 1. Then the integral over this part of the path is

$$\int_{z=0}^{1} \left(3(1)^2+6(1)\right)0 \;-\; 14(1)\,z(0) \;+\; 20(1)\,z^2\,dz \;=\; \int_{z=0}^{1} 20z^2\,dz \;=\; \frac{20z^3}{3}\Big|_0^1 \;=\; \frac{20}{3}$$

Adding,
$$\int_C \mathbf{A}\cdot d\mathbf{r} \;=\; 1 + 0 + \frac{20}{3} \;=\; \frac{23}{3}$$

(c) The straight line joining (0,0,0) and (1,1,1) is given in parametric form by $x=t$, $y=t$, $z=t$. Then

$$\int_C \mathbf{A}\cdot d\mathbf{r} \;=\; \int_{t=0}^{1} (3t^2+6t)\,dt \;-\; 14(t)(t)\,dt \;+\; 20(t)(t)^2\,dt$$

$$= \int_{t=0}^{1} (3t^2+6t-14t^2+20t^3)\,dt \;=\; \int_{t=0}^{1} (6t-11t^2+20t^3)\,dt \;=\; \frac{13}{3}$$

7. Find the total work done in moving a particle in a force field given by $\mathbf{F} = 3xy\,\mathbf{i} - 5z\,\mathbf{j} + 10x\,\mathbf{k}$ along the curve $x=t^2+1$, $y=2t^2$, $z=t^3$ from $t=1$ to $t=2$.

$$\text{Total work} \;=\; \int_C \mathbf{F}\cdot d\mathbf{r} \;=\; \int_C (3xy\,\mathbf{i} - 5z\,\mathbf{j} + 10x\,\mathbf{k})\cdot(dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k})$$

$$= \int_C 3xy\,dx \;-\; 5z\,dy \;+\; 10x\,dz$$

$$= \int_{t=1}^{2} 3(t^2+1)(2t^2)\,d(t^2+1) \;-\; 5(t^3)\,d(2t^2) \;+\; 10(t^2+1)\,d(t^3)$$

$$= \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2)\,dt \;=\; 303$$

8. If $\mathbf{F} = 3xy\,\mathbf{i} - y^2\,\mathbf{j}$, evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ where $C$ is the curve in the $xy$ plane, $y=2x^2$, from (0,0) to (1,2).

Since the integration is performed in the $xy$ plane ($z=0$), we can take $\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j}$. Then

$$\int_C \mathbf{F}\cdot d\mathbf{r} \;=\; \int_C (3xy\,\mathbf{i} - y^2\,\mathbf{j})\cdot(dx\,\mathbf{i} + dy\,\mathbf{j})$$

$$= \int_C 3xy\,dx \;-\; y^2\,dy$$

*First Method.* Let $x=t$ in $y=2x^2$. Then the parametric equations of $C$ are $x=t$, $y=2t^2$. Points (0,0) and (1,2) correspond to $t=0$ and $t=1$ respectively. Then

$$\int_C \mathbf{F}\cdot d\mathbf{r} \;=\; \int_{t=0}^{1} 3(t)(2t^2)\,dt \;-\; (2t^2)^2\,d(2t^2) \;=\; \int_{t=0}^{1} (6t^3-16t^5)\,dt \;=\; -\frac{7}{6}$$

*Second Method.* Substitute $y=2x^2$ directly, where $x$ goes from 0 to 1. Then

$$\int_C \mathbf{F}\cdot d\mathbf{r} \;=\; \int_{x=0}^{1} 3x(2x^2)\,dx \;-\; (2x^2)^2\,d(2x^2) \;=\; \int_{x=0}^{1} (6x^3-16x^5)\,dx \;=\; -\frac{7}{6}$$

Note that if the curve were traversed in the opposite sense, i.e. from (1,2) to (0,0), the value of the integral would have been 7/6 instead of $-7/6$.

**9.** Find the work done in moving a particle once around a circle $C$ in the $xy$ plane, if the circle has center at the origin and radius 3 and if the force field is given by

$$\mathbf{F} = (2x - y + z)\mathbf{i} + (x + y - z^2)\mathbf{j} + (3x - 2y + 4z)\mathbf{k}$$

In the plane $z=0$, $\mathbf{F} = (2x-y)\mathbf{i} + (x+y)\mathbf{j} + (3x-2y)\mathbf{k}$ and $d\mathbf{r} = dx\,\mathbf{i} + dy\,\mathbf{j}$ so that the work done is

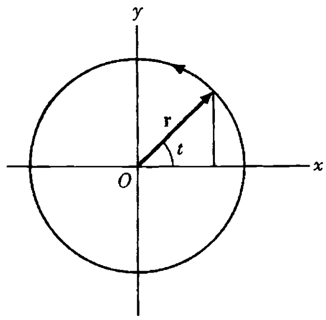
$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C [(2x-y)\mathbf{i} + (x+y)\mathbf{j} + (3x-2y)\mathbf{k}]\cdot[dx\,\mathbf{i} + dy\,\mathbf{j}]$$

$$= \int_C (2x-y)\,dx + (x+y)\,dy$$

Choose the parametric equations of the circle as $x = 3\cos t$, $y = 3\sin t$ where $t$ varies from 0 to $2\pi$ (see adjoining figure). Then the line integral equals

$$\int_{t=0}^{2\pi} [2(3\cos t) - 3\sin t][-3\sin t]\,dt + [3\cos t + 3\sin t][3\cos t]\,dt$$

$$= \int_0^{2\pi} (9 - 9\sin t\cos t)\,dt = 9t - \frac{9}{2}\sin^2 t\Big|_0^{2\pi} = 18\pi$$




$\mathbf{r} = x\mathbf{i} + y\mathbf{j}$
$= 3\cos t\,\mathbf{i} + 3\sin t\,\mathbf{j}$

In traversing $C$ we have chosen the counterclockwise direction indicated in the adjoining figure. We call this the *positive* direction, or say that $C$ has been traversed in the positive sense. If $C$ were traversed in the clockwise (negative) direction the value of the integral would be $-18\pi$.

**10.** (*a*) If $\mathbf{F} = \nabla\phi$, where $\phi$ is single-valued and has continuous partial derivatives, show that the work done in moving a particle from one point $P_1 \equiv (x_1, y_1, z_1)$ in this field to another point $P_2 \equiv (x_2, y_2, z_2)$ is independent of the path joining the two points.

(*b*) Conversely, if $\int_C \mathbf{F}\cdot d\mathbf{r}$ is independent of the path $C$ joining any two points, show that there exists a function $\phi$ such that $\mathbf{F} = \nabla\phi$.

(*a*) Work done $= \int_{P_1}^{P_2} \mathbf{F}\cdot d\mathbf{r} = \int_{P_1}^{P_2} \nabla\phi\cdot d\mathbf{r}$

$$= \int_{P_1}^{P_2} \left(\frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}\right)\cdot(dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k})$$

$$= \int_{P_1}^{P_2} \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz$$

$$= \int_{P_1}^{P_2} d\phi = \phi(P_2) - \phi(P_1) = \phi(x_2, y_2, z_2) - \phi(x_1, y_1, z_1)$$

Then the integral depends only on points $P_1$ and $P_2$ and not on the path joining them. This is true of course only if $\phi(x,y,z)$ is single-valued at all points $P_1$ and $P_2$.

(*b*) Let $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$. By hypothesis, $\int_C \mathbf{F}\cdot d\mathbf{r}$ is independent of the path $C$ joining any two points, which we take as $(x_1, y_1, z_1)$ and $(x,y,z)$ respectively. Then

$$\phi(x,y,z) = \int_{(x_1,y_1,z_1)}^{(x,y,z)} \mathbf{F}\cdot d\mathbf{r} = \int_{(x_1,y_1,z_1)}^{(x,y,z)} F_1\,dx + F_2\,dy + F_3\,dz$$

is independent of the path joining $(x_1, y_1, z_1)$ and $(x,y,z)$. Thus

$$\phi(x+\Delta x, y, z) - \phi(x,y,z) = \int_{(x_1,y_1,z_1)}^{(x+\Delta x,y,z)} \mathbf{F}\cdot d\mathbf{r} - \int_{(x_1,y_1,z_1)}^{(x,y,z)} \mathbf{F}\cdot d\mathbf{r}$$

$$= \int_{(x,y,z)}^{(x_1,y_1,z_1)} \mathbf{F}\cdot d\mathbf{r} + \int_{(x_1,y_1,z_1)}^{(x+\Delta x,y,z)} \mathbf{F}\cdot d\mathbf{r}$$

$$= \int_{(x,y,z)}^{(x+\Delta x,y,z)} \mathbf{F}\cdot d\mathbf{r} = \int_{(x,y,z)}^{(x+\Delta x,y,z)} F_1\,dx + F_2\,dy + F_3\,dz$$

Since the last integral must be independent of the path joining $(x,y,z)$ and $(x+\Delta x,y,z)$, we may choose the path to be a straight line joining these points so that $dy$ and $dz$ are zero. Then

$$\frac{\phi(x+\Delta x, y, z) - \phi(x,y,z)}{\Delta x} = \frac{1}{\Delta x}\int_{(x,y,z)}^{(x+\Delta x,y,z)} F_1\,dx$$

Taking the limit of both sides as $\Delta x \to 0$, we have $\dfrac{\partial\phi}{\partial x} = F_1$.

Similarly, we can show that $\dfrac{\partial\phi}{\partial y} = F_2$ and $\dfrac{\partial\phi}{\partial z} = F_3$.

Then $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k} = \dfrac{\partial\phi}{\partial x}\mathbf{i} + \dfrac{\partial\phi}{\partial y}\mathbf{j} + \dfrac{\partial\phi}{\partial z}\mathbf{k} = \nabla\phi$.

If $\displaystyle\int_{P_1}^{P_2} \mathbf{F}\cdot d\mathbf{r}$ is independent of the path $C$ joining $P_1$ and $P_2$, then $\mathbf{F}$ is called a *conservative field.* It follows that if $\mathbf{F} = \nabla\phi$ then $\mathbf{F}$ is conservative, and conversely.

*Proof using vectors.* If the line integral is independent of the path, then

$$\phi(x,y,z) = \int_{(x_1,y_1,z_1)}^{(x,y,z)} \mathbf{F}\cdot d\mathbf{r} = \int_{(x_1,y_1,z_1)}^{(x,y,z)} \mathbf{F}\cdot\frac{d\mathbf{r}}{ds}\,ds$$

By differentiation, $\dfrac{d\phi}{ds} = \mathbf{F}\cdot\dfrac{d\mathbf{r}}{ds}$. But $\dfrac{d\phi}{ds} = \nabla\phi\cdot\dfrac{d\mathbf{r}}{ds}$ so that $(\nabla\phi - \mathbf{F})\cdot\dfrac{d\mathbf{r}}{ds} = 0$.

Since this must hold irrespective of $\dfrac{d\mathbf{r}}{ds}$, we have $\mathbf{F} = \nabla\phi$.

11. (*a*) If $\mathbf{F}$ is a conservative field, prove that $\operatorname{curl}\mathbf{F} = \nabla\times\mathbf{F} = \mathbf{0}$ (i.e. $\mathbf{F}$ is irrotational).
(*b*) Conversely, if $\nabla\times\mathbf{F} = \mathbf{0}$ (i.e. $\mathbf{F}$ is irrotational), prove that $\mathbf{F}$ is conservative.

(*a*) If $\mathbf{F}$ is a conservative field, then by Problem 10, $\mathbf{F} = \nabla\phi$.
Thus $\operatorname{curl}\mathbf{F} = \nabla\times\nabla\phi = \mathbf{0}$ (see Problem 27(*a*), Chapter 4).

(*b*) If $\nabla\times\mathbf{F} = \mathbf{0}$, then $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \mathbf{0}$ and thus

$$\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z}, \qquad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}, \qquad \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$$

We must prove that $\mathbf{F} = \nabla\phi$ follows as a consequence of this.

The work done in moving a particle from $(x_1,y_1,z_1)$ to $(x,y,z)$ in the force field $\mathbf{F}$ is

$$\int_C F_1(x,y,z)\,dx + F_2(x,y,z)\,dy + F_3(x,y,z)\,dz$$

where $C$ is a path joining $(x_1,y_1,z_1)$ and $(x,y,z)$. Let us choose as a particular path the straight line segments from $(x_1,y_1,z_1)$ to $(x,y_1,z_1)$ to $(x,y,z_1)$ to $(x,y,z)$ and call $\phi(x,y,z)$ the work done along this particular path. Then

$$\phi(x,y,z) = \int_{x_1}^{x} F_1(x,y_1,z_1)\,dx + \int_{y_1}^{y} F_2(x,y,z_1)\,dy + \int_{z_1}^{z} F_3(x,y,z)\,dz$$

It follows that

$$\frac{\partial\phi}{\partial z} = F_3(x,y,z)$$

$$\begin{aligned}\frac{\partial\phi}{\partial y} &= F_2(x,y,z_1) + \int_{z_1}^{z} \frac{\partial F_3}{\partial y}(x,y,z)\,dz \\ &= F_2(x,y,z_1) + \int_{z_1}^{z} \frac{\partial F_2}{\partial z}(x,y,z)\,dz \\ &= F_2(x,y,z_1) + F_2(x,y,z)\Big|_{z_1}^{z} = F_2(x,y,z_1) + F_2(x,y,z) - F_2(x,y,z_1) = F_2(x,y,z)\end{aligned}$$

$$\begin{aligned}\frac{\partial\phi}{\partial x} &= F_1(x,y_1,z_1) + \int_{y_1}^{y} \frac{\partial F_2}{\partial x}(x,y,z_1)\,dy + \int_{z_1}^{z} \frac{\partial F_3}{\partial x}(x,y,z)\,dz \\ &= F_1(x,y_1,z_1) + \int_{y_1}^{y} \frac{\partial F_1}{\partial y}(x,y,z_1)\,dy + \int_{z_1}^{z} \frac{\partial F_1}{\partial z}(x,y,z)\,dz \\ &= F_1(x,y_1,z_1) + F_1(x,y,z_1)\Big|_{y_1}^{y} + F_1(x,y,z)\Big|_{z_1}^{z} \\ &= F_1(x,y_1,z_1) + F_1(x,y,z_1) - F_1(x,y_1,z_1) + F_1(x,y,z) - F_1(x,y,z_1) = F_1(x,y,z)\end{aligned}$$

Then $\quad \mathbf{F} = F_1\,\mathbf{i} + F_2\,\mathbf{j} + F_3\,\mathbf{k} = \dfrac{\partial\phi}{\partial x}\mathbf{i} + \dfrac{\partial\phi}{\partial y}\mathbf{j} + \dfrac{\partial\phi}{\partial z}\mathbf{k} = \nabla\phi.$

Thus a necessary and sufficient condition that a field $\mathbf{F}$ be conservative is that $\operatorname{curl}\mathbf{F} = \nabla\times\mathbf{F} = \mathbf{0}$.

**12.** (*a*) Show that $\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$ is a conservative force field. (*b*) Find the scalar potential. (*c*) Find the work done in moving an object in this field from $(1,-2,1)$ to $(3,1,4)$.

(*a*) From Problem 11, a necessary and sufficient condition that a force will be conservative is that $\operatorname{curl}\mathbf{F} = \nabla\times\mathbf{F} = \mathbf{0}$.

Now $\quad \nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix} = \mathbf{0}.$

Thus $\mathbf{F}$ is a conservative force field.

(*b*) *First Method.*

By Problem 10, $\mathbf{F} = \nabla\phi$ or $\frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k} = (2xy+z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$. Then

$$(1)\ \frac{\partial\phi}{\partial x} = 2xy + z^3 \qquad (2)\ \frac{\partial\phi}{\partial y} = x^2 \qquad (3)\ \frac{\partial\phi}{\partial z} = 3xz^2$$

Integrating, we find from (*1*), (*2*) and (*3*) respectively,

$$\begin{aligned} \phi &= x^2y + xz^3 + f(y,z) \\ \phi &= x^2y + g(x,z) \\ \phi &= xz^3 + h(x,y) \end{aligned}$$

These agree if we choose $f(y,z) = 0$, $g(x,z) = xz^3$, $h(x,y) = x^2y$ so that $\phi = x^2y + xz^3$ to which may be added any constant.

*Second Method.*

Since $\mathbf{F}$ is conservative, $\int_C \mathbf{F}\cdot d\mathbf{r}$ is independent of the path $C$ joining $(x_1,y_1,z_1)$ and $(x,y,z)$. Using the method of Problem 11(*b*),

$$\begin{aligned} \phi(x,y,z) &= \int_{x_1}^{x} (2xy_1 + z_1^3)\,dx + \int_{y_1}^{y} x^2\,dy + \int_{z_1}^{z} 3xz^2\,dz \\ &= (x^2y_1 + xz_1^3)\Big|_{x_1}^{x} + x^2y\Big|_{y_1}^{y} + xz^3\Big|_{z_1}^{z} \\ &= x^2y_1 + xz_1^3 - x_1^2y_1 - x_1z_1^3 + x^2y - x^2y_1 + xz^3 - xz_1^3 \\ &= x^2y + xz^3 - x_1^2y_1 - x_1z_1^3 = x^2y + xz^3 + \text{constant} \end{aligned}$$

*Third Method.* $\mathbf{F}\cdot d\mathbf{r} = \nabla\phi\cdot d\mathbf{r} = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = d\phi$

Then

$$\begin{aligned} d\phi = \mathbf{F}\cdot d\mathbf{r} &= (2xy+z^3)\,dx + x^2\,dy + 3xz^2\,dz \\ &= (2xy\,dx + x^2\,dy) + (z^3\,dx + 3xz^2\,dz) \\ &= d(x^2y) + d(xz^3) = d(x^2y + xz^3) \end{aligned}$$

and $\phi = x^2y + xz^3 + \text{constant}$.

(*c*) Work done

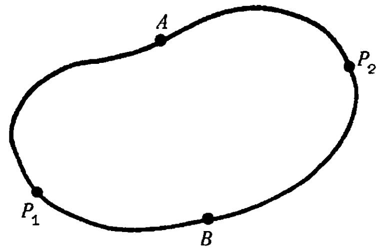
$$\begin{aligned} &= \int_{P_1}^{P_2} \mathbf{F}\cdot d\mathbf{r} \\ &= \int_{P_1}^{P_2} (2xy+z^3)\,dx + x^2\,dy + 3xz^2\,dz \\ &= \int_{P_1}^{P_2} d(x^2y + xz^3) = x^2y + xz^3\Big|_{P_1}^{P_2} = x^2y + xz^3\Big|_{(1,-2,1)}^{(3,1,4)} = 202 \end{aligned}$$

*Another Method.*

From part (*b*), $\phi(x,y,z) = x^2y + xz^3 + \text{constant}$.

Then work done $= \phi(3,1,4) - \phi(1,-2,1) = 202$.

**13.** Prove that if $\displaystyle\int_{P_1}^{P_2} \mathbf{F}\cdot d\mathbf{r}$ is independent of the path joining any two points $P_1$ and $P_2$ in a given region, then $\displaystyle\oint \mathbf{F}\cdot d\mathbf{r} = 0$ for all closed paths in the region and conversely.




Let $P_1AP_2BP_1$ (see adjacent figure) be a closed curve. Then

$$\oint \mathbf{F}\cdot d\mathbf{r} = \int_{P_1AP_2BP_1} \mathbf{F}\cdot d\mathbf{r} = \int_{P_1AP_2} \mathbf{F}\cdot d\mathbf{r} + \int_{P_2BP_1} \mathbf{F}\cdot d\mathbf{r}$$
$$= \int_{P_1AP_2} \mathbf{F}\cdot d\mathbf{r} - \int_{P_1BP_2} \mathbf{F}\cdot d\mathbf{r} = 0$$

since the integral from $P_1$ to $P_2$ along a path through $A$ is the same as that along a path through $B$, by hypothesis.

Conversely if $\oint \mathbf{F}\cdot d\mathbf{r} = 0$, then

$$\int_{P_1AP_2BP_1} \mathbf{F}\cdot d\mathbf{r} = \int_{P_1AP_2} \mathbf{F}\cdot d\mathbf{r} + \int_{P_2BP_1} \mathbf{F}\cdot d\mathbf{r} = \int_{P_1AP_2} \mathbf{F}\cdot d\mathbf{r} - \int_{P_1BP_2} \mathbf{F}\cdot d\mathbf{r} = 0$$

so that, $\displaystyle\int_{P_1AP_2} \mathbf{F}\cdot d\mathbf{r} = \int_{P_1BP_2} \mathbf{F}\cdot d\mathbf{r}$.

**14.** (*a*) Show that a necessary and sufficient condition that $F_1\,dx + F_2\,dy + F_3\,dz$ be an exact differential is that $\nabla\times\mathbf{F} = \mathbf{0}$ where $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$.

(*b*) Show that $(y^2z^3\cos x - 4x^3z)\,dx + 2z^3y\sin x\,dy + (3y^2z^2\sin x - x^4)\,dz$ is an exact differential of a function $\phi$ and find $\phi$.

(*a*) Suppose $F_1\,dx + F_2\,dy + F_3\,dz = d\phi = \dfrac{\partial\phi}{\partial x}dx + \dfrac{\partial\phi}{\partial y}dy + \dfrac{\partial\phi}{\partial z}dz$, an exact differential. Then since $x, y$ and $z$ are independent variables,

$$F_1 = \frac{\partial\phi}{\partial x}, \quad F_2 = \frac{\partial\phi}{\partial y}, \quad F_3 = \frac{\partial\phi}{\partial z}$$

and so $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k} = \dfrac{\partial\phi}{\partial x}\mathbf{i} + \dfrac{\partial\phi}{\partial y}\mathbf{j} + \dfrac{\partial\phi}{\partial z}\mathbf{k} = \nabla\phi$. Thus $\nabla\times\mathbf{F} = \nabla\times\nabla\phi = \mathbf{0}$.

Conversely if $\nabla\times\mathbf{F} = \mathbf{0}$ then by Problem 11, $\mathbf{F} = \nabla\phi$ and so $\mathbf{F}\cdot d\mathbf{r} = \nabla\phi\cdot d\mathbf{r} = d\phi$, i.e. $F_1\,dx + F_2\,dy + F_3\,dz = d\phi$, an exact differential.

(*b*) $\mathbf{F} = (y^2z^3\cos x - 4x^3z)\mathbf{i} + 2z^3y\sin x\,\mathbf{j} + (3y^2z^2\sin x - x^4)\mathbf{k}$ and $\nabla\times\mathbf{F}$ is computed to be zero, so that by part (*a*)

$$(y^2z^3\cos x - 4x^3z)\,dx + 2z^3y\sin x\,dy + (3y^2z^2\sin x - x^4)\,dz = d\phi$$

By any of the methods of Problem 12 we find $\phi = y^2z^3\sin x - x^4z + \text{constant}$.

**15.** Let $\mathbf{F}$ be a conservative force field such that $\mathbf{F} = -\nabla\phi$. Suppose a particle of constant mass $m$ to move in this field. If $A$ and $B$ are any two points in space, prove that

$$\phi(A) + \tfrac{1}{2}mv_A^2 = \phi(B) + \tfrac{1}{2}mv_B^2$$

where $v_A$ and $v_B$ are the magnitudes of the velocities of the particle at $A$ and $B$ respectively.

$\mathbf{F} = m\mathbf{a} = m\dfrac{d^2\mathbf{r}}{dt^2}$. Then $\mathbf{F}\cdot\dfrac{d\mathbf{r}}{dt} = m\dfrac{d\mathbf{r}}{dt}\cdot\dfrac{d^2\mathbf{r}}{dt^2} = \dfrac{m}{2}\dfrac{d}{dt}\left(\dfrac{d\mathbf{r}}{dt}\right)^2$.

Integrating, $\displaystyle\int_A^B \mathbf{F}\cdot d\mathbf{r} = \frac{m}{2}v^2\Big|_A^B = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2$.

If $\mathbf{F} = -\nabla\phi$, $\displaystyle\int_A^B \mathbf{F}\cdot d\mathbf{r} = -\int_A^B \nabla\phi\cdot d\mathbf{r} = -\int_A^B d\phi = \phi(A) - \phi(B)$.

Then $\phi(A) - \phi(B) = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2$ and the result follows.

$\phi(A)$ is called the *potential energy* at $A$ and $\frac{1}{2}mv_A^2$ is the *kinetic energy* at $A$. The result states that the total energy at $A$ equals the total energy at $B$ (conservation of energy). Note the use of the minus sign in $\mathbf{F} = -\nabla\phi$.

**16.** If $\phi = 2xyz^2$, $\mathbf{F} = xy\,\mathbf{i} - z\,\mathbf{j} + x^2\,\mathbf{k}$ and $C$ is the curve $x=t^2$, $y=2t$, $z=t^3$ from $t=0$ to $t=1$, evaluate the line integrals (a) $\displaystyle\int_C \phi\, d\mathbf{r}$, (b) $\displaystyle\int_C \mathbf{F}\times d\mathbf{r}$.

(a) Along $C$,

$$\phi = 2xyz^2 = 2(t^2)(2t)(t^3)^2 = 4t^9,$$
$$\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k} = t^2\,\mathbf{i} + 2t\,\mathbf{j} + t^3\,\mathbf{k}, \quad \text{and}$$
$$d\mathbf{r} = (2t\,\mathbf{i} + 2\mathbf{j} + 3t^2\,\mathbf{k})\,dt. \quad \text{Then}$$

$$\int_C \phi\, d\mathbf{r} = \int_{t=0}^1 4t^9(2t\,\mathbf{i} + 2\mathbf{j} + 3t^2\,\mathbf{k})\,dt$$
$$= \mathbf{i}\int_0^1 8t^{10}\,dt + \mathbf{j}\int_0^1 8t^9\,dt + \mathbf{k}\int_0^1 12t^{11}\,dt = \frac{8}{11}\mathbf{i} + \frac{4}{5}\mathbf{j} + \mathbf{k}$$

(b) Along $C$, $\mathbf{F} = xy\,\mathbf{i} - z\,\mathbf{j} + x^2\,\mathbf{k} = 2t^3\,\mathbf{i} - t^3\,\mathbf{j} + t^4\,\mathbf{k}$.

Then
$$\mathbf{F}\times d\mathbf{r} = (2t^3\,\mathbf{i} - t^3\,\mathbf{j} + t^4\,\mathbf{k}) \times (2t\,\mathbf{i} + 2\mathbf{j} + 3t^2\,\mathbf{k})\,dt$$
$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2t^3 & -t^3 & t^4 \\ 2t & 2 & 3t^2 \end{vmatrix} dt = \left[(-3t^5 - 2t^4)\mathbf{i} + (2t^5 - 6t^5)\mathbf{j} + (4t^3 + 2t^4)\mathbf{k}\right] dt$$

and
$$\int_C \mathbf{F}\times d\mathbf{r} = \mathbf{i}\int_0^1 (-3t^5 - 2t^4)\,dt + \mathbf{j}\int_0^1 (-4t^5)\,dt + \mathbf{k}\int_0^1 (4t^3 + 2t^4)\,dt$$
$$= -\frac{9}{10}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{7}{5}\mathbf{k}$$

## SURFACE INTEGRALS.

**17.** Give a definition of $\displaystyle\iint_S \mathbf{A}\cdot\mathbf{n}\, dS$ over a surface $S$ in terms of limit of a sum.

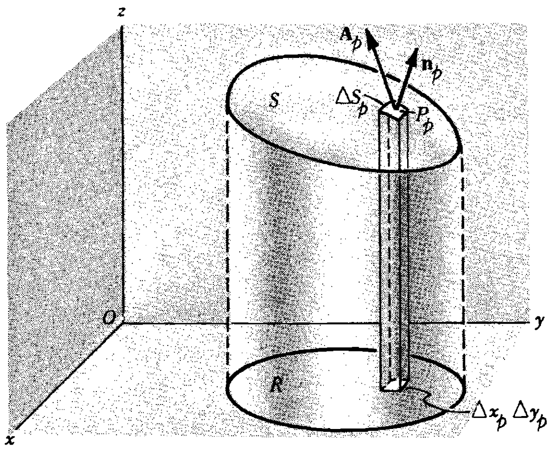
Subdivide the area $S$ into $M$ elements of area $\Delta S_p$ where $p = 1,2,3,\dots,M$. Choose any point $P_p$ within $\Delta S_p$ whose coordinates are $(x_p, y_p, z_p)$. Define $\mathbf{A}(x_p, y_p, z_p) = \mathbf{A}_p$. Let $\mathbf{n}_p$ be the positive unit normal to $\Delta S_p$ at $P$. Form the sum

$$\sum_{p=1}^{M} \mathbf{A}_p \cdot \mathbf{n}_p \, \Delta S_p$$

where $\mathbf{A}_p \cdot \mathbf{n}_p$ is the normal component of $\mathbf{A}_p$ at $P_p$.

Now take the limit of this sum as $M \to \infty$ in such a way that the largest dimension of each $\Delta S_p$ approaches zero. This limit, if it exists, is called the surface integral of the normal component of $\mathbf{A}$ over $S$ and is denoted by

$$\iint_S \mathbf{A} \cdot \mathbf{n} \, dS$$




**18.** Suppose that the surface $S$ has projection $R$ on the $xy$ plane (see figure of Prob.17). Show that

$$\iint_S \mathbf{A} \cdot \mathbf{n} \, dS \quad = \quad \iint_R \mathbf{A} \cdot \mathbf{n} \frac{dx\,dy}{|\mathbf{n} \cdot \mathbf{k}|}$$

By Problem 17, the surface integral is the limit of the sum

$$(1) \qquad \sum_{p=1}^{M} \mathbf{A}_p \cdot \mathbf{n}_p \, \Delta S_p$$

The projection of $\Delta S_p$ on the $xy$ plane is $|(\mathbf{n}_p \, \Delta S_p) \cdot \mathbf{k}|$ or $|\mathbf{n}_p \cdot \mathbf{k}| \, \Delta S_p$ which is equal to $\Delta x_p \Delta y_p$ so that $\Delta S_p = \dfrac{\Delta x_p \Delta y_p}{|\mathbf{n}_p \cdot \mathbf{k}|}$. Thus the sum (1) becomes

$$(2) \qquad \sum_{p=1}^{M} \mathbf{A}_p \cdot \mathbf{n}_p \frac{\Delta x_p \Delta y_p}{|\mathbf{n}_p \cdot \mathbf{k}|}$$

By the fundamental theorem of integral calculus the limit of this sum as $M \to \infty$ in such a manner that the largest $\Delta x_p$ and $\Delta y_p$ approach zero is

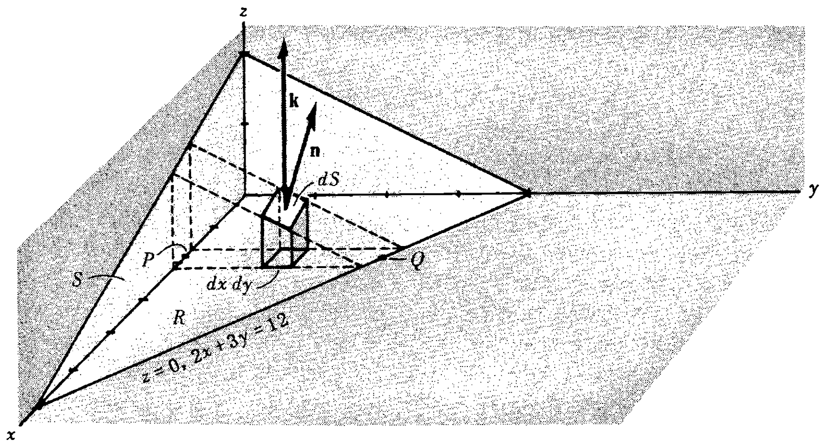
$$\iint_R \mathbf{A} \cdot \mathbf{n} \frac{dx\,dy}{|\mathbf{n} \cdot \mathbf{k}|}$$

and so the required result follows.

Strictly speaking, the result $\Delta S_p = \dfrac{\Delta x_p \Delta y_p}{|\mathbf{n}_p \cdot \mathbf{k}|}$ is only approximately true but it can be shown on closer examination that they differ from each other by infinitesimals of order higher than $\Delta x_p \Delta y_p$, and using this the *limits* of (1) and (2) can in fact be shown equal.

**19.** Evaluate $\displaystyle\iint_S \mathbf{A} \cdot \mathbf{n} \, dS$, where $\mathbf{A} = 18z\,\mathbf{i} - 12\mathbf{j} + 3y\,\mathbf{k}$ and $S$ is that part of the plane $2x + 3y + 6z = 12$ which is located in the first octant.

The surface $S$ and its projection $R$ on the $xy$ plane are shown in the figure below.




From Problem 17,

$$\iint_S \mathbf{A}\cdot\mathbf{n}\,dS = \iint_R \mathbf{A}\cdot\mathbf{n}\,\frac{dx\,dy}{|\mathbf{n}\cdot\mathbf{k}|}$$

To obtain $\mathbf{n}$ note that a vector perpendicular to the surface $2x+3y+6z = 12$ is given by $\nabla(2x+3y+6z) = 2\mathbf{i}+3\mathbf{j}+6\mathbf{k}$ (see Problem 5 of Chapter 4). Then a unit normal to any point of $S$ (see figure above) is

$$\mathbf{n} = \frac{2\mathbf{i}+3\mathbf{j}+6\mathbf{k}}{\sqrt{2^2+3^2+6^2}} = \tfrac{2}{7}\mathbf{i} + \tfrac{3}{7}\mathbf{j} + \tfrac{6}{7}\mathbf{k}$$

Thus $\mathbf{n}\cdot\mathbf{k} = (\tfrac{2}{7}\mathbf{i} + \tfrac{3}{7}\mathbf{j} + \tfrac{6}{7}\mathbf{k})\cdot\mathbf{k} = \tfrac{6}{7}$ and so $\dfrac{dx\,dy}{|\mathbf{n}\cdot\mathbf{k}|} = \tfrac{7}{6}\,dx\,dy$.

Also $\mathbf{A}\cdot\mathbf{n} = (18z\,\mathbf{i} - 12\mathbf{j} + 3y\,\mathbf{k})\cdot(\tfrac{2}{7}\mathbf{i}+\tfrac{3}{7}\mathbf{j}+\tfrac{6}{7}\mathbf{k}) = \dfrac{36z - 36 + 18y}{7} = \dfrac{36-12x}{7}$, using the fact that $z = \dfrac{12-2x-3y}{6}$ from the equation of $S$. Then

$$\iint_S \mathbf{A}\cdot\mathbf{n}\,dS = \iint_R \mathbf{A}\cdot\mathbf{n}\,\frac{dx\,dy}{|\mathbf{n}\cdot\mathbf{k}|} = \iint_R \left(\frac{36-12x}{7}\right)\frac{7}{6}\,dx\,dy = \iint_R (6-2x)\,dx\,dy$$

To evaluate this double integral over $R$, keep $x$ fixed and integrate with respect to $y$ from $y=0$ ($P$ in the figure above) to $y = \dfrac{12-2x}{3}$ ($Q$ in the figure above); then integrate with respect to $x$ from $x=0$ to $x=6$. In this manner $R$ is completely covered. The integral becomes

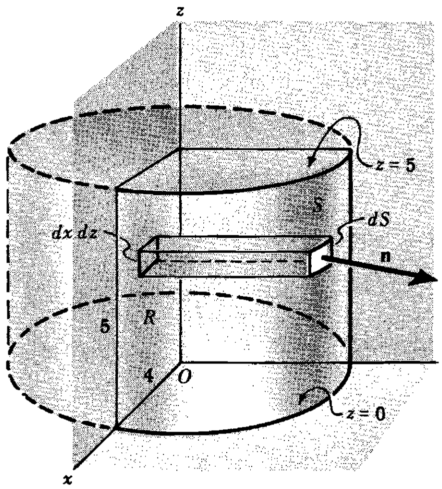
$$\int_{x=0}^{6}\int_{y=0}^{(12-2x)/3} (6-2x)\,dy\,dx = \int_{x=0}^{6} (24 - 12x + \tfrac{4x^2}{3})\,dx = 24$$

If we had chosen the positive unit normal $\mathbf{n}$ opposite to that in the figure above, we would have obtained the result $-24$.

**20.** Evaluate $\iint_S \mathbf{A}\cdot\mathbf{n}\,dS$, where $\mathbf{A} = z\,\mathbf{i} + x\,\mathbf{j} - 3y^2z\,\mathbf{k}$ and $S$ is the surface of the cylinder $x^2+y^2 = 16$ included in the first octant between $z=0$ and $z=5$.

Project $S$ on the $xz$ plane as in the figure below and call the projection $R$. Note that the projection of $S$ on the $xy$ plane cannot be used here. Then

$$\iint_S \mathbf{A}\cdot\mathbf{n}\,dS = \iint_R \mathbf{A}\cdot\mathbf{n}\,\frac{dx\,dz}{|\mathbf{n}\cdot\mathbf{j}|}$$




A normal to $x^2+y^2=16$ is $\nabla(x^2+y^2) = 2x\mathbf{i}+2y\mathbf{j}$. Thus the unit normal to $S$ as shown in the adjoining figure, is

$$\mathbf{n} = \frac{2x\,\mathbf{i}+2y\,\mathbf{j}}{\sqrt{(2x)^2+(2y)^2}} = \frac{x\,\mathbf{i}+y\,\mathbf{j}}{4}$$

since $x^2+y^2=16$ on $S$.

$$\mathbf{A}\cdot\mathbf{n} = (z\,\mathbf{i}+x\,\mathbf{j}-3y^2z\,\mathbf{k})\cdot\left(\frac{x\,\mathbf{i}+y\,\mathbf{j}}{4}\right) = \frac{1}{4}(xz+xy)$$

$$\mathbf{n}\cdot\mathbf{j} = \frac{x\,\mathbf{i}+y\,\mathbf{j}}{4}\cdot\mathbf{j} = \frac{y}{4}.$$

Then the surface integral equals

$$\iint_R \frac{xz+xy}{y}\,dx\,dz = \int_{z=0}^{5}\int_{x=0}^{4}\left(\frac{xz}{\sqrt{16-x^2}}+x\right)dx\,dz = \int_{z=0}^{5}(4z+8)\,dz = 90$$

**21.** Evaluate $\iint_S \phi\mathbf{n}\,dS$ where $\phi = \frac{3}{8}xyz$ and $S$ is the surface of Problem 20.

We have
$$\iint_S \phi\mathbf{n}\,dS = \iint_R \phi\mathbf{n}\,\frac{dx\,dz}{|\mathbf{n}\cdot\mathbf{j}|}$$

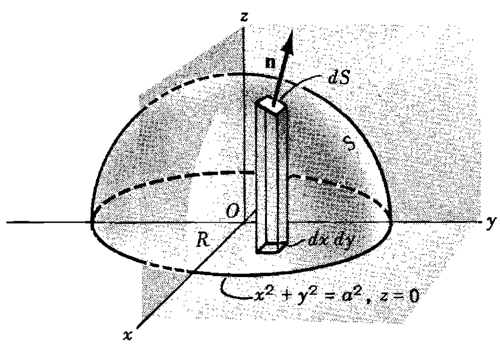
Using $\mathbf{n} = \frac{x\,\mathbf{i}+y\,\mathbf{j}}{4}$, $\mathbf{n}\cdot\mathbf{j} = \frac{y}{4}$ as in Problem 20, this last integral becomes

$$\iint_R \frac{3}{8}xz\,(x\,\mathbf{i}+y\,\mathbf{j})\,dx\,dz = \frac{3}{8}\int_{z=0}^{5}\int_{x=0}^{4}(x^2z\,\mathbf{i}+xz\sqrt{16-x^2}\,\mathbf{j})\,dx\,dz$$

$$= \frac{3}{8}\int_{z=0}^{5}\left(\frac{64}{3}z\,\mathbf{i}+\frac{64}{3}z\,\mathbf{j}\right)dz = 100\,\mathbf{i} + 100\,\mathbf{j}$$

**22.** If $\mathbf{F} = y\,\mathbf{i}+(x-2xz)\,\mathbf{j}-xy\,\mathbf{k}$, evaluate $\iint_S (\nabla\times\mathbf{F})\cdot\mathbf{n}\,dS$ where $S$ is the surface of the sphere $x^2+y^2+z^2=a^2$ above the $xy$ plane.

$$\nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & x-2xz & -xy \end{vmatrix} = x\,\mathbf{i}+y\,\mathbf{j}-2z\,\mathbf{k}$$




A normal to $x^2+y^2+z^2=a^2$ is

$$\nabla(x^2+y^2+z^2) = 2x\,\mathbf{i}+2y\,\mathbf{j}+2z\,\mathbf{k}$$

Then the unit normal $\mathbf{n}$ of the figure above is given by

$$\mathbf{n} = \frac{2x\,\mathbf{i} + 2y\,\mathbf{j} + 2z\,\mathbf{k}}{\sqrt{4x^2+4y^2+4z^2}} = \frac{x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}}{a}$$

since $x^2 + y^2 + z^2 = a^2$.

The projection of $S$ on the $xy$ plane is the region $R$ bounded by the circle $x^2+y^2 = a^2$, $z=0$ (see figure above). Then

$$\iint_S (\nabla\times\mathbf{F})\cdot\mathbf{n}\,dS = \iint_R (\nabla\times\mathbf{F})\cdot\mathbf{n}\,\frac{dx\,dy}{|\mathbf{n}\cdot\mathbf{k}|}$$

$$= \iint_R (x\,\mathbf{i} + y\,\mathbf{j} - 2z\,\mathbf{k})\cdot\left(\frac{x\,\mathbf{i}+y\,\mathbf{j}+z\,\mathbf{k}}{a}\right)\frac{dx\,dy}{z/a}$$

$$= \int_{x=-a}^{a}\int_{y=-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \frac{3(x^2+y^2) - 2a^2}{\sqrt{a^2-x^2-y^2}}\,dy\,dx$$

using the fact that $z = \sqrt{a^2-x^2-y^2}$. To evaluate the double integral, transform to polar coordinates $(\rho,\phi)$ where $x = \rho\cos\phi$, $y = \rho\sin\phi$ and $dy\,dx$ is replaced by $\rho\,d\rho\,d\phi$. The double integral becomes

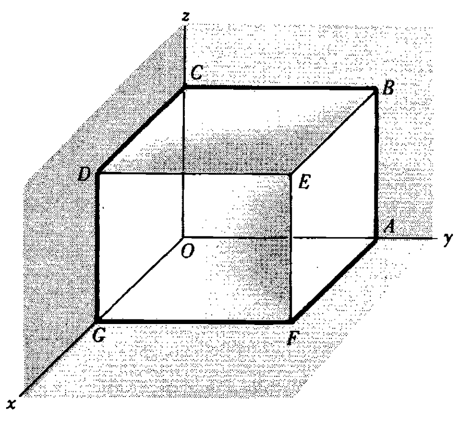
$$\int_{\phi=0}^{2\pi}\int_{\rho=0}^{a} \frac{3\rho^2 - 2a^2}{\sqrt{a^2-\rho^2}}\,\rho\,d\rho\,d\phi = \int_{\phi=0}^{2\pi}\int_{\rho=0}^{a} \frac{3(\rho^2-a^2)+a^2}{\sqrt{a^2-\rho^2}}\,\rho\,d\rho\,d\phi$$

$$= \int_{\phi=0}^{2\pi}\int_{\rho=0}^{a} \left(-3\rho\sqrt{a^2-\rho^2} + \frac{a^2\rho}{\sqrt{a^2-\rho^2}}\right) d\rho\,d\phi$$

$$= \int_{\phi=0}^{2\pi} \left[(a^2-\rho^2)^{3/2} - a^2\sqrt{a^2-\rho^2}\,\Big|_{\rho=0}^{a}\right] d\phi$$

$$= \int_{\phi=0}^{2\pi} (a^3 - a^3)\,d\phi = 0$$

23. If $\mathbf{F} = 4xz\,\mathbf{i} - y^2\,\mathbf{j} + yz\,\mathbf{k}$, evaluate $\iint_S \mathbf{F}\cdot\mathbf{n}\,dS$ where $S$ is the surface of the cube bounded by $x=0$, $x=1$, $y=0$, $y=1$, $z=0$, $z=1$.




*Face DEFG:* $\mathbf{n}=\mathbf{i}$, $x=1$. Then

$$\iint_{DEFG} \mathbf{F}\cdot\mathbf{n}\,dS = \int_0^1\int_0^1 (4z\,\mathbf{i} - y^2\,\mathbf{j} + yz\,\mathbf{k})\cdot\mathbf{i}\,dy\,dz$$

$$= \int_0^1\int_0^1 4z\,dy\,dz = 2$$

*Face ABCO*: $\mathbf{n} = -\mathbf{i}$, $x = 0$. Then

$$\iint_{ABCO} \mathbf{F}\cdot\mathbf{n}\,dS = \int_0^1\int_0^1 (-y^2\,\mathbf{j} + yz\,\mathbf{k})\cdot(-\mathbf{i})\,dy\,dz = 0$$

*Face ABEF*: $\mathbf{n} = \mathbf{j}$, $y = 1$. Then

$$\iint_{ABEF} \mathbf{F}\cdot\mathbf{n}\,dS = \int_0^1\int_0^1 (4xz\,\mathbf{i} - \mathbf{j} + z\,\mathbf{k})\cdot\mathbf{j}\,dx\,dz = \int_0^1\int_0^1 -dx\,dz = -1$$

*Face OGDC*: $\mathbf{n} = -\mathbf{j}$, $y = 0$. Then

$$\iint_{OGDC} \mathbf{F}\cdot\mathbf{n}\,dS = \int_0^1\int_0^1 (4xz\,\mathbf{i})\cdot(-\mathbf{j})\,dx\,dz = 0$$

*Face BCDE*: $\mathbf{n} = \mathbf{k}$, $z = 1$. Then

$$\iint_{BCDE} \mathbf{F}\cdot\mathbf{n}\,dS = \int_0^1\int_0^1 (4x\,\mathbf{i} - y^2\,\mathbf{j} + y\,\mathbf{k})\cdot\mathbf{k}\,dx\,dy = \int_0^1\int_0^1 y\,dx\,dy = \tfrac{1}{2}$$

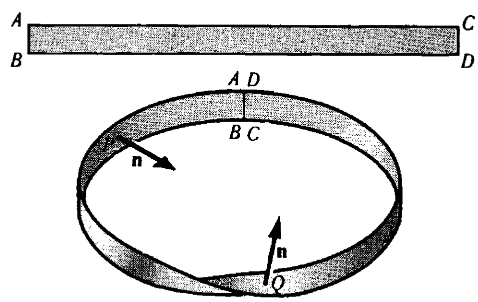
*Face AFGO*: $\mathbf{n} = -\mathbf{k}$, $z = 0$. Then

$$\iint_{AFGO} \mathbf{F}\cdot\mathbf{n}\,dS = \int_0^1\int_0^1 (-y^2\,\mathbf{j})\cdot(-\mathbf{k})\,dx\,dy = 0$$

Adding, $\displaystyle\iint_S \mathbf{F}\cdot\mathbf{n}\,dS = 2 + 0 + (-1) + 0 + \tfrac{1}{2} + 0 = \frac{3}{2}$.

**24.** In dealing with surface integrals we have restricted ourselves to surfaces which are two-sided. Give an example of a surface which is not two-sided.

Take a strip of paper such as $ABCD$ as shown in the adjoining figure. Twist the strip so that points $A$ and $B$ fall on $D$ and $C$ respectively, as in the adjoining figure. If $\mathbf{n}$ is the positive normal at point $P$ of the surface, we find that as $\mathbf{n}$ moves around the surface it reverses its original direction when it reaches $P$ again. If we tried to color only one side of the surface we would find the whole thing colored. This surface, called a *Moebius strip*, is an example of a one-sided surface. This is sometimes called a *non-orientable* surface. A two-sided surface is *orientable*.



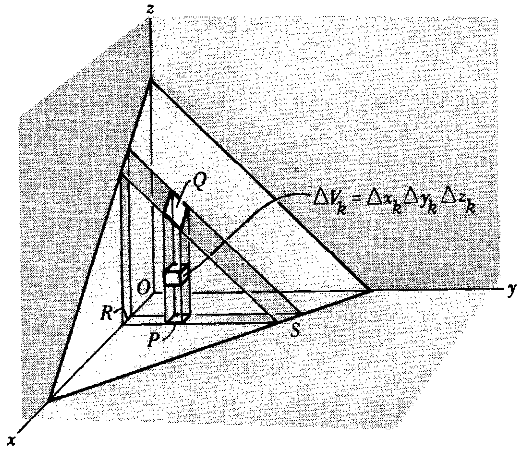

## VOLUME INTEGRALS

**25.** Let $\phi = 45x^2y$ and let $V$ denote the closed region bounded by the planes $4x + 2y + z = 8$, $x = 0$, $y = 0$, $z = 0$. (*a*) Express $\displaystyle\iiint_V \phi\,dV$ as the limit of a sum. (*b*) Evaluate the integral in (*a*).

(a) Subdivide region $V$ into $M$ cubes having volume $\Delta V_k = \Delta x_k \Delta y_k \Delta z_k$ $k = 1, 2, \ldots, M$ as indicated in the adjoining figure and let $(x_k, y_k, z_k)$ be a point within this cube. Define $\phi(x_k, y_k, z_k) = \phi_k$. Consider the sum

$$(1) \qquad \sum_{k=1}^{M} \phi_k \Delta V_k$$

taken over all possible cubes in the region. The limit of this sum, when $M \to \infty$ in such a manner that the largest of the quantities $\Delta V_k$ will approach zero, if it exists, is denoted by $\iiint_V \phi \, dV$. It can be shown that this limit is independent of the method of subdivision if $\phi$ is continuous throughout $V$.




In forming the sum (1) over all possible cubes in the region, it is advisable to proceed in an orderly fashion. One possibility is to add first all terms in (1) corresponding to volume elements contained in a column such as $PQ$ in the above figure. This amounts to keeping $x_k$ and $y_k$ fixed and adding over all $z_k$'s. Next, keep $x_k$ fixed but sum over all $y_k$'s. This amounts to adding all columns such as $PQ$ contained in a slab $RS$, and consequently amounts to summing over all cubes contained in such a slab. Finally, vary $x_k$. This amounts to addition of all slabs such as $RS$.

In the process outlined the summation is taken first over $z_k$'s then over $y_k$'s and finally over $x_k$'s. However, the summation can clearly be taken in any other order.

(b) The ideas involved in the method of summation outlined in (a) can be used in evaluating the integral. Keeping $x$ and $y$ constant, integrate from $z = 0$ (base of column $PQ$) to $z = 8 - 4x - 2y$ (top of column $PQ$). Next keep $x$ constant and integrate with respect to $y$. This amounts to addition of columns having bases in the $xy$ plane ($z = 0$) located anywhere from $R$ (where $y = 0$) to $S$ (where $4x+2y = 8$ or $y = 4 - 2x$), and the integration is from $y = 0$ to $y = 4 - 2x$. Finally, we add all slabs parallel to the $yz$ plane, which amounts to integration from $x = 0$ to $x = 2$. The integration can be written

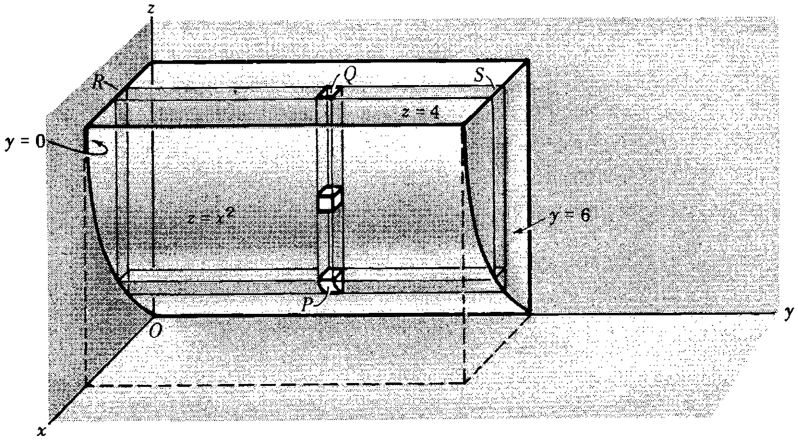
$$\int_{x=0}^{2} \int_{y=0}^{4-2x} \int_{z=0}^{8-4x-2y} 45x^2 y \, dz\, dy\, dx \;=\; 45 \int_{x=0}^{2} \int_{y=0}^{4-2x} x^2 y (8 - 4x - 2y)\, dy\, dx$$

$$= \; 45 \int_{x=0}^{2} \frac{1}{3} x^2 (4 - 2x)^3 \, dx \;=\; 128$$

Note: Physically the result can be interpreted as the mass of the region $V$ in which the density $\phi$ varies according to the formula $\phi = 45x^2 y$.

**26.** Let $\mathbf{F} = 2xz\,\mathbf{i} - x\,\mathbf{j} + y^2\mathbf{k}$. Evaluate $\iiint_V \mathbf{F}\, dV$ where $V$ is the region bounded by the surfaces $x = 0$, $y = 0$, $y = 6$, $z = x^2$, $z = 4$.

The region $V$ is covered (a) by keeping $x$ and $y$ fixed and integrating from $z = x^2$ to $z = 4$ (base to top of column $PQ$), (b) then by keeping $x$ fixed and integrating from $y = 0$ to $y = 6$ ($R$ to $S$ in the slab), (c) finally integrating from $x = 0$ to $x = 2$ (where $z = x^2$ meets $z = 4$). Then the required integral is

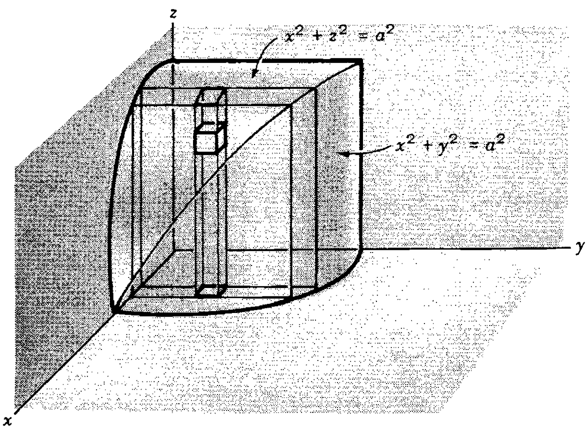



$$\int_{x=0}^{2}\int_{y=0}^{6}\int_{z=x^2}^{4}(2xz\,\mathbf{i} - x\,\mathbf{j} + y^2\,\mathbf{k})\,dz\,dy\,dx$$

$$= \mathbf{i}\int_0^2\int_0^6\int_{x^2}^4 2xz\,dz\,dy\,dx \;-\; \mathbf{j}\int_0^2\int_0^6\int_{x^2}^4 x\,dz\,dy\,dx \;+\; \mathbf{k}\int_0^2\int_0^6\int_{x^2}^4 y^2\,dz\,dy\,dx$$

$$= 128\,\mathbf{i} - 24\,\mathbf{j} + 384\,\mathbf{k}$$

**27.** Find the volume of the region common to the intersecting cylinders $x^2+y^2 = a^2$ and $x^2+z^2 = a^2$.




Required volume = 8 times volume of region shown in above figure

$$= 8\int_{x=0}^{a}\int_{y=0}^{\sqrt{a^2-x^2}}\int_{z=0}^{\sqrt{a^2-x^2}} dz\,dy\,dx$$

$$= 8\int_{x=0}^{a}\int_{y=0}^{\sqrt{a^2-x^2}} \sqrt{a^2-x^2}\,dy\,dx = 8\int_{x=0}^{a}(a^2-x^2)\,dx = \frac{16a^3}{3}$$

## SUPPLEMENTARY PROBLEMS

28. If $\mathbf{R}(t) = (3t^2-t)\mathbf{i} + (2-6t)\mathbf{j} - 4t\,\mathbf{k}$, find (a) $\int \mathbf{R}(t)\,dt$ and (b) $\int_2^4 \mathbf{R}(t)\,dt$.

*Ans.* (a) $(t^3-t^2/2)\mathbf{i} + (2t-3t^2)\mathbf{j} - 2t^2\mathbf{k} + \mathbf{c}$ (b) $50\mathbf{i} - 32\mathbf{j} - 24\mathbf{k}$

29. Evaluate $\int_0^{\pi/2} (3\sin u\,\mathbf{i} + 2\cos u\,\mathbf{j})\,du$ *Ans.* $3\mathbf{i} + 2\mathbf{j}$

30. If $\mathbf{A}(t) = t\,\mathbf{i} - t^2\mathbf{j} + (t-1)\mathbf{k}$ and $\mathbf{B}(t) = 2t^2\mathbf{i} + 6t\,\mathbf{k}$, evaluate (a) $\int_0^2 \mathbf{A}\cdot\mathbf{B}\,dt$, (b) $\int_0^2 \mathbf{A}\times\mathbf{B}\,dt$.

*Ans.* (a) 12 (b) $-24\mathbf{i} - \frac{40}{3}\mathbf{j} + \frac{64}{5}\mathbf{k}$

31. Let $\mathbf{A} = t\,\mathbf{i} - 3\mathbf{j} + 2t\,\mathbf{k}$, $\mathbf{B} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$, $\mathbf{C} = 3\mathbf{i} + t\,\mathbf{j} - \mathbf{k}$. Evaluate (a) $\int_1^2 \mathbf{A}\cdot\mathbf{B}\times\mathbf{C}\,dt$, (b) $\int_1^2 \mathbf{A}\times(\mathbf{B}\times\mathbf{C})\,dt$.

*Ans.* (a) 0 (b) $-\frac{87}{2}\mathbf{i} - \frac{44}{3}\mathbf{j} + \frac{15}{2}\mathbf{k}$

32. The acceleration $\mathbf{a}$ of a particle at any time $t \geqq 0$ is given by $\mathbf{a} = e^{-t}\mathbf{i} - 6(t+1)\mathbf{j} + 3\sin t\,\mathbf{k}$. If the velocity $\mathbf{v}$ and displacement $\mathbf{r}$ are zero at $t=0$, find $\mathbf{v}$ and $\mathbf{r}$ at any time.

*Ans.* $\mathbf{v} = (1-e^{-t})\mathbf{i} - (3t^2+6t)\mathbf{j} + (3-3\cos t)\mathbf{k}$, $\mathbf{r} = (t-1+e^{-t})\mathbf{i} - (t^3+3t^2)\mathbf{j} + (3t-3\sin t)\mathbf{k}$

33. The acceleration $\mathbf{a}$ of an object at any time $t$ is given by $\mathbf{a} = -g\mathbf{j}$, where $g$ is a constant. At $t=0$ the velocity is given by $\mathbf{v} = v_0\cos\theta_0\,\mathbf{i} + v_0\sin\theta_0\,\mathbf{j}$ and the displacement $\mathbf{r} = \mathbf{0}$. Find $\mathbf{v}$ and $\mathbf{r}$ at any time $t>0$. This describes the motion of a projectile fired from a cannon inclined at angle $\theta_0$ with the positive $x$-axis with initial velocity of magnitude $v_0$.

*Ans.* $\mathbf{v} = v_0\cos\theta_0\,\mathbf{i} + (v_0\sin\theta_0 - gt)\mathbf{j}$, $\mathbf{r} = (v_0\cos\theta_0)t\,\mathbf{i} + [(v_0\sin\theta_0)t - \frac{1}{2}gt^2]\mathbf{j}$

34. Evaluate $\int_2^3 \mathbf{A}\cdot\frac{d\mathbf{A}}{dt}\,dt$ if $\mathbf{A}(2) = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{A}(3) = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$. *Ans.* 10

35. Find the areal velocity of a particle which moves along the path $\mathbf{r} = a\cos\omega t\,\mathbf{i} + b\sin\omega t\,\mathbf{j}$ where $a,b,\omega$ are constants and $t$ is time. *Ans.* $\frac{1}{2}ab\omega\,\mathbf{k}$

36. Prove that the squares of the periods of planets in their motion around the sun are proportional to the cubes of the major axes of their elliptical paths (Kepler's third law).

37. If $\mathbf{A} = (2y+3)\mathbf{i} + xz\,\mathbf{j} + (yz-x)\mathbf{k}$, evaluate $\int_C \mathbf{A}\cdot d\mathbf{r}$ along the following paths $C$:

(a) $x=2t^2$, $y=t$, $z=t^3$ from $t=0$ to $t=1$,
(b) the straight lines from (0,0,0) to (0,0,1), then to (0,1,1), and then to (2,1,1),
(c) the straight line joining (0,0,0) and (2,1,1).

*Ans.* (a) 288/35 (b) 10 (c) 8

38. If $\mathbf{F} = (5xy-6x^2)\mathbf{i} + (2y-4x)\mathbf{j}$, evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ along the curve $C$ in the $xy$ plane, $y=x^3$ from the point (1,1) to (2,8). *Ans.* 35

39. If $\mathbf{F} = (2x+y)\mathbf{i} + (3y-x)\mathbf{j}$, evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ where $C$ is the curve in the $xy$ plane consisting of the straight lines from (0,0) to (2,0) and then to (3,2). *Ans.* 11

40. Find the work done in moving a particle in the force field $\mathbf{F} = 3x^2\mathbf{i} + (2xz-y)\mathbf{j} + z\,\mathbf{k}$ along

(a) the straight line from (0,0,0) to (2,1,3).
(b) the space curve $x=2t^2$, $y=t$, $z=4t^2-t$ from $t=0$ to $t=1$.
(c) the curve defined by $x^2=4y$, $3x^3=8z$ from $x=0$ to $x=2$.

*Ans.* (a) 16 (b) 14.2 (c) 16

41. Evaluate $\oint_C \mathbf{F}\cdot d\mathbf{r}$ where $\mathbf{F} = (x-3y)\mathbf{i} + (y-2x)\mathbf{j}$ and $C$ is the closed curve in the $xy$ plane, $x = 2\cos t$, $y = 3\sin t$ from $t=0$ to $t=2\pi$. *Ans.* $6\pi$, if $C$ is traversed in the positive (counterclockwise) direction.

42. If $\mathbf{T}$ is a unit tangent vector to the curve $C$, $\mathbf{r}=\mathbf{r}(u)$, show that the work done in moving a particle in a force field $\mathbf{F}$ along $C$ is given by $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$ where $s$ is the arc length.

43. If $\mathbf{F} = (2x+y^2)\mathbf{i} + (3y-4x)\mathbf{j}$, evaluate $\oint_C \mathbf{F}\cdot d\mathbf{r}$ around the triangle $C$ of Figure 1, (*a*) in the indicated direction, (*b*) opposite to the indicated direction. *Ans.* (*a*) $-14/3$ (*b*) $14/3$

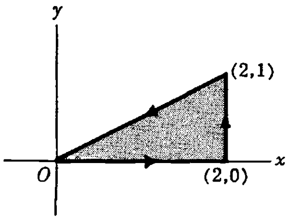



Fig. 1

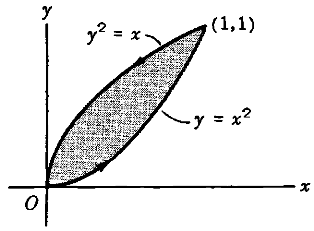



Fig. 2

44. Evaluate $\oint_C \mathbf{A}\cdot d\mathbf{r}$ around the closed curve $C$ of Fig. 2 above if $\mathbf{A} = (x-y)\mathbf{i} + (x+y)\mathbf{j}$. *Ans.* 2/3

45. If $\mathbf{A} = (y-2x)\mathbf{i} + (3x+2y)\mathbf{j}$, compute the circulation of $\mathbf{A}$ about a circle $C$ in the $xy$ plane with center at the origin and radius 2, if $C$ is traversed in the positive direction. *Ans.* $8\pi$

46. (*a*) If $\mathbf{A} = (4xy-3x^2z^2)\mathbf{i} + 2x^2\mathbf{j} - 2x^3z\,\mathbf{k}$, prove that $\int_C \mathbf{A}\cdot d\mathbf{r}$ is independent of the curve $C$ joining two given points. (*b*) Show that there is a differentiable function $\phi$ such that $\mathbf{A} = \nabla\phi$ and find it.
*Ans.* (*b*) $\phi = 2x^2y - x^3z^2 + \text{constant}$

47. (*a*) Prove that $\mathbf{F} = (y^2\cos x + z^3)\mathbf{i} + (2y\sin x - 4)\mathbf{j} + (3xz^2+2)\mathbf{k}$ is a conservative force field.
(*b*) Find the scalar potential for $\mathbf{F}$.
(*c*) Find the work done in moving an object in this field from $(0,1,-1)$ to $(\pi/2,-1,2)$.
*Ans.* (*b*) $\phi = y^2\sin x + xz^3 - 4y + 2z + \text{constant}$ (*c*) $15 + 4\pi$

48. Prove that $\mathbf{F} = r^2\mathbf{r}$ is conservative and find the scalar potential. *Ans.* $\phi = \frac{r^4}{4} + \text{constant}$

49. Determine whether the force field $\mathbf{F} = 2xz\,\mathbf{i} + (x^2-y)\mathbf{j} + (2z-x^2)\mathbf{k}$ is conservative or non-conservative.
*Ans.* non-conservative

50. Show that the work done on a particle in moving it from $A$ to $B$ equals its change in kinetic energies at these points whether the force field is conservative or not.

51. Evaluate $\int_C \mathbf{A}\cdot d\mathbf{r}$ along the curve $x^2+y^2=1$, $z=1$ in the positive direction from $(0,1,1)$ to $(1,0,1)$ if $\mathbf{A} = (yz+2x)\mathbf{i} + xz\,\mathbf{j} + (xy+2z)\mathbf{k}$. *Ans.* 1

52. (*a*) If $\mathbf{E} = r\mathbf{r}$, is there a function $\phi$ such that $\mathbf{E} = -\nabla\phi$? If so, find it. (*b*) Evaluate $\oint_C \mathbf{E}\cdot d\mathbf{r}$ if $C$ is any simple closed curve. *Ans.* (*a*) $\phi = -\frac{r^3}{3} + \text{constant}$ (*b*) 0

53. Show that $(2x\cos y + z\sin y)\,dx + (xz\cos y - x^2\sin y)\,dy + x\sin y\,dz$ is an exact differential. Hence

solve the differential equation $(2x\cos y + z\sin y)\,dx + (xz\cos y - x^2\sin y)\,dy + x\sin y\,dz = 0$.
*Ans.* $x^2\cos y + xz\sin y = \text{constant}$

**54.** Solve (*a*) $(e^{-y} + 3x^2y^2)\,dx + (2x^3y - xe^{-y})\,dy = 0$,
(*b*) $(z - e^{-x}\sin y)\,dx + (1 + e^{-x}\cos y)\,dy + (x - 8z)\,dz = 0$.
*Ans.* (*a*) $xe^{-y} + x^3y^2 = \text{constant}$ (*b*) $xz + e^{-x}\sin y + y - 4z^2 = \text{constant}$

**55.** If $\phi = 2xy^2z + x^2y$, evaluate $\int_C \phi\,d\mathbf{r}$ where $C$
(*a*) is the curve $x = t,\ y = t^2,\ z = t^3$ from $t = 0$ to $t = 1$
(*b*) consists of the straight lines from (0,0,0) to (1,0,0), then to (1,1,0), and then to (1,1,1).
*Ans.* (*a*) $\frac{19}{45}\mathbf{i} + \frac{11}{15}\mathbf{j} + \frac{75}{77}\mathbf{k}$ (*b*) $\frac{1}{2}\mathbf{j} + 2\mathbf{k}$

**56.** If $\mathbf{F} = 2y\,\mathbf{i} - z\,\mathbf{j} + x\,\mathbf{k}$, evaluate $\int_C \mathbf{F}\times d\mathbf{r}$ along the curve $x = \cos t,\ y = \sin t,\ z = 2\cos t$ from $t = 0$ to $t = \pi/2$. *Ans.* $(2 - \frac{\pi}{4})\mathbf{i} + (\pi - \frac{1}{2})\mathbf{j}$

**57.** If $\mathbf{A} = (3x+y)\mathbf{i} - x\,\mathbf{j} + (y-2)\mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, evaluate $\oint_C (\mathbf{A}\times\mathbf{B})\times d\mathbf{r}$ around the circle in the $xy$ plane having center at the origin and radius 2 traversed in the positive direction. *Ans.* $4\pi(7\mathbf{i} + 3\mathbf{j})$

**58.** Evaluate $\iint_S \mathbf{A}\cdot\mathbf{n}\,dS$ for each of the following cases.
(*a*) $\mathbf{A} = y\,\mathbf{i} + 2x\,\mathbf{j} - z\,\mathbf{k}$ and $S$ is the surface of the plane $2x + y = 6$ in the first octant cut off by the plane $z = 4$.
(*b*) $\mathbf{A} = (x + y^2)\mathbf{i} - 2x\,\mathbf{j} + 2yz\,\mathbf{k}$ and $S$ is the surface of the plane $2x + y + 2z = 6$ in the first octant.
*Ans.* (*a*) 108 (*b*) 81

**59.** If $\mathbf{F} = 2y\,\mathbf{i} - z\,\mathbf{j} + x^2\,\mathbf{k}$ and $S$ is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$, evaluate $\iint_S \mathbf{F}\cdot\mathbf{n}\,dS$. *Ans.* 132

**60.** Evaluate $\iint_S \mathbf{A}\cdot\mathbf{n}\,dS$ over the entire surface $S$ of the region bounded by the cylinder $x^2 + z^2 = 9$, $x = 0$, $y = 0$, $z = 0$ and $y = 8$, if $\mathbf{A} = 6z\,\mathbf{i} + (2x + y)\mathbf{j} - x\,\mathbf{k}$. *Ans.* $18\pi$

**61.** Evaluate $\iint_S \mathbf{r}\cdot\mathbf{n}\,dS$ over: (*a*) the surface $S$ of the unit cube bounded by the coordinate planes and the planes $x = 1$, $y = 1$, $z = 1$; (*b*) the surface of a sphere of radius $a$ with center at (0,0,0).
*Ans.* (*a*) 3 (*b*) $4\pi a^3$

**62.** Evaluate $\iint_S \mathbf{A}\cdot\mathbf{n}\,dS$ over the entire surface of the region above the $xy$ plane bounded by the cone $z^2 = x^2 + y^2$ and the plane $z = 4$, if $\mathbf{A} = 4xz\,\mathbf{i} + xyz^2\,\mathbf{j} + 3z\,\mathbf{k}$. *Ans.* $320\pi$

**63.** (*a*) Let $R$ be the projection of a surface $S$ on the $xy$ plane. Prove that the surface area of $S$ is given by $\iint_R \sqrt{1 + (\frac{\partial z}{\partial x})^2 + (\frac{\partial z}{\partial y})^2}\,dx\,dy$ if the equation for $S$ is $z = f(x,y)$.

(*b*) What is the surface area if $S$ has the equation $F(x,y,z)=0$? *Ans.* $\displaystyle\iint_R \frac{\sqrt{(\frac{\partial F}{\partial x})^2+(\frac{\partial F}{\partial y})^2+(\frac{\partial F}{\partial z})^2}}{\left|\frac{\partial F}{\partial z}\right|}\,dx\,dy$

**64.** Find the surface area of the plane $x+2y+2z=12$ cut off by: (*a*) $x=0$, $y=0$, $x=1$, $y=1$; (*b*) $x=0$, $y=0$, and $x^2+y^2=16$. *Ans.* (*a*) 3/2 (*b*) $6\pi$

**65.** Find the surface area of the region common to the intersecting cylinders $x^2+y^2=a^2$ and $x^2+z^2=a^2$. *Ans.* $16a^2$

**66.** Evaluate (*a*) $\displaystyle\iint_S (\nabla\times\mathbf{F})\cdot\mathbf{n}\,dS$ and (*b*) $\displaystyle\iint_S \phi\,\mathbf{n}\,dS$ if $\mathbf{F}=(x+2y)\mathbf{i}-3z\,\mathbf{j}+x\,\mathbf{k}$, $\phi=4x+3y-2z$, and $S$ is the surface of $2x+y+2z=6$ bounded by $x=0$, $x=1$, $y=0$ and $y=2$.
*Ans.* (*a*) 1 (*b*) $2\mathbf{i}+\mathbf{j}+2\mathbf{k}$

**67.** Solve the preceding problem if $S$ is the surface of $2x+y+2z=6$ bounded by $x=0$, $y=0$, and $z=0$.
*Ans.* (*a*) 9/2 (*b*) $72\mathbf{i}+36\mathbf{j}+72\mathbf{k}$

**68.** Evaluate $\displaystyle\iint_R \sqrt{x^2+y^2}\,dx\,dy$ over the region $R$ in the $xy$ plane bounded by $x^2+y^2=36$. *Ans.* $144\pi$

**69.** Evaluate $\displaystyle\iiint_V (2x+y)\,dV$, where $V$ is the closed region bounded by the cylinder $z=4-x^2$ and the planes $x=0$, $y=0$, $y=2$ and $z=0$. *Ans.* 80/3

**70.** If $\mathbf{F}=(2x^2-3z)\mathbf{i}-2xy\,\mathbf{j}-4x\,\mathbf{k}$, evaluate (*a*) $\displaystyle\iiint_V \nabla\cdot\mathbf{F}\,dV$ and (*b*) $\displaystyle\iiint_V \nabla\times\mathbf{F}\,dV$, where $V$ is the closed region bounded by the planes $x=0$, $y=0$, $z=0$ and $2x+2y+z=4$. *Ans.* (*a*) $\frac{8}{3}$ (*b*) $\frac{8}{3}(\mathbf{j}-\mathbf{k})$